

When Reputation Backfires: Social Pressure and Contribution Withdrawal in Multi-Seed LLM Climate Societies

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Abstract

Reputation and gossip are widely treated as stabilizers of cooperation in multi-agent societies. We test that assumption for large language model (LLM) agents in a climate-risk public-goods economy with asymmetric vulnerability and a dual-use loss-and-damage pool. Two independent full-protocol runs, identical except for the random seed, each place twenty-six agents in thirty rounds under forced institutional assignment, climatic shocks, democratic parameter updating, peer consistency scoring, aggregate reputation, and gossip bulletins. The primary outcome is contribution intensity relative to wealth. Across both runs, agents in the non-enforcing institution reduce intensity after bad-reputation and gossip events on average. Responses among enforcing-institution agents reverse sign across seeds. Average cooperation remains moderately positive in both societies, yet path stability is limited and democratic adaptation is path dependent. Shock damage coverage is high while developed–developing wealth gaps widen. The robust pattern is institutional heterogeneity in social pressure. Social monitoring associates with withdrawal among developing agents without Stage-2 enforcement, rather than with universal reputation repair.

Keywords: large language model agents; multi-agent systems; reputation; gossip; cooperation; climate risk sharing; endogenous institutions; public goods; multi-seed robustness

1. Introduction

Collective climate action couples a public-goods dilemma with unequal vulnerability and contested redistribution. Agents must decide how much of private wealth to transfer when peers can observe, gossip about, and sanction behavior, while formal rules may change through voting. Classical theory treats reputation as a cooperation technology. Visible giving raises standing. Standing protects future partners and disciplines free-riding (Milinski et al. 2002; Wedekind and Milinski 2000; Nowak 2006). That directional claim now travels into societies of large language model (LLM) agents, where peer talk and scoring are easy to scaffold (Park et al. 2023; Piatti et al. 2024; Ren et al. 2025).

Unfortunately, LLM agents also exhibit calculated free-riding once payoff frames dominate deliberation (Li and Shirado 2025; Guzman Piedrahita et al. 2025). Communication and monitoring can therefore fail as repair devices even when average transfers look cooperative. Climate asymmetry sharpens the test. Developing agents may lack peer-enforcement tools. Social stigma may land without a credible repair path. Forced institutional assignment removes exit as an escape valve that earlier free-choice baselines allow.

In contrast, we treat social pressure as an event to be measured rather than as an assumed stabilizer. We embed Meta Llama 3.1 8B agents in a repeated climate-risk economy with two parallel institutions, climatic shocks, a dual-use loss-and-damage fund, democratic whitelist voting, pairwise consistency scoring, aggregate reputation, and gossip bulletins. Contribution intensity relative to wealth is the primary outcome. We event-study intensity after bad-reputation marks, reputation drops, and gossip targeting. We code post-event rationales for opportunistic versus repair motifs.

A single realization cannot separate structural behavior from stochastic path dependence. Consequently, we replicate the full protocol under two independent seeds and report both runs with equal weight. The multi-seed design yields a sharper claim than a one-shot narrative. Across seeds, mean intensity after social marks falls for agents in the non-enforcing institution. For agents in the enforcing institution, the gossip association flips from negative to positive. Average cooperation levels differ across seeds, yet institution means remain matched within each seed. Democracy expands subsidies in both runs, while punishment weakening is adopted in only one.

Our contributions are fourfold. First, we provide a multi-seed event study of reputation and gossip under forced climate institutions. Second, we separate robust withdrawal among non-enforcing agents from seed-sensitive responses among enforcing agents. Third, we show that positive average cooperation coexists with limited path stability and sparse reputation-repair language. Fourth, we document coverage without equalization in the loss-and-damage pool across both fiscal trajectories. Together, these results challenge the default claim that social monitoring raises cooperation in LLM climate societies.

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They also show why single-run narratives are insufficient for social-mechanism claims in stochastic LLM societies.

2. Literature Review

2.1 Collective action, sanctions, and reputation

Collective-action problems arise when private incentives erode shared returns (Olson 1965; Hardin 1968). Laboratory and evolutionary traditions show that monitoring, costly punishment, and reputation can sustain cooperation (Fehr and Gächter 2002; Boyd et al. 2003; Axelrod and Hamilton 1981; Axelrod 1984). Indirect reciprocity and image scoring formalize how observed giving shapes future partners (Wedekind and Milinski 2000; Panchanathan and Boyd 2004). Milinski and colleagues argue that reputation helps solve commons dilemmas by linking contribution to standing (Milinski et al. 2002). Nowak lists reputation among the core evolutionary routes to cooperation (Nowak 2006). Across these lines, the working hypothesis is directional. More social visibility should raise, or at least protect, cooperative transfers.

Institutional design complements reputation. Ostrom documents polycentric rules, graduated sanctions, and participatory governance in durable commons regimes (Ostrom 1990). Those design principles motivate dual channels in computational societies. Agents may face formal enforcement inside one institution and social evaluation everywhere. They may also revise fiscal parameters by vote. The classical literatures rarely ask what happens when social marks land on agents who cannot exit and cannot punish peers.

2.2 Language-model agents as strategic laboratories

Language-model agents change the measurement surface of social dilemmas. They produce numeric choices together with natural-language rationales, belief updates, and peer judgments (Park et al. 2023; Sumers et al. 2024). Game-theoretic LLM platforms treat such agents as laboratories for strategic interaction (Mao et al. 2023). Cooperation benchmarks report heterogeneous success across games, prompts, and communication regimes (Piatti et al. 2024; Madmoun and Lahlou 2025; Liang et al. 2025). Some societies sustain contributions when talk is rich. Others collapse once incentives dominate.

A growing strand shows that explicit reasoning can reduce cooperation. Models that deliberate more intensely often favor individually rational free-riding (Li and Shirado 2025; Guzman Piedrahita et al. 2025). That pattern mirrors human “spontaneous giving and calculated greed” and warns against equating fluent justification with prosocial commitment. Norm-focused multi-agent studies likewise report partial emergence and fragile stability rather than automatic internalization (Gupta et al. 2025; Ashery et al. 2025; Cross et al. 2025). These results motivate separating average transfer levels from norm strength.

2.3 Reputation systems as assumed stabilizers

Several recent systems treat reputation networks or peer talk as cooperation technology. RepuNet-style architectures encode standing so agents can condition on peer history (Ren et al. 2025). GovSim and related communication protocols emphasize information sharing as a coordination route (Piatti

et al. 2024; Madmoun and Lahlou 2025). Punishment-strategy transplants from classical evolutionary models show that explicit sanctions can organize behavior when costs and targets are clear (Warnakulasuriya et al. 2025). The cumulative message is optimistic. If agents can talk and score one another, collective outcomes should improve.

That optimism is incomplete for climate asymmetry. Where gossip substitutes for formal punishment, the mapping from stigma to repair is unsettled. Self-reinforcing salience loops in LLM inference can lock opportunistic frames once they appear (Drake 2025). Identification of free-riders without sanction reach is a known failure mode in mixed populations. Social monitoring can then name agents without raising contributions. Prior LLM work rarely event-studies contribution intensity after negative social marks under forced institutional partitions.

2.4 Climate risk sharing and computational governance

Climate governance couples mitigation public goods with asymmetric vulnerability and compensation claims. Loss-and-damage instruments are dual-use in spirit. Contributions can fund shared returns and contingent payouts after shocks. Computational societies that combine those fiscal rules with LLM agents that gossip and vote remain rare. Endogenous democracy lets agents revise punishment, subsidy, and equity weights. That channel is politically meaningful. It also confounds clean shock identification when votes coincide with damage rounds.

Inequality complicates interpretation of social pressure (Lu et al. 2025; Ashery et al. 2025). If stigma falls on low-capacity agents without Stage-2 tools, withdrawal may be a constrained response. Shock-free free-choice baselines in related work show that reputation can coordinate sorting when exit is available. They also show identification of defectors without cross-institution sanctioning power. Those boundary results define when reputation helps versus when it coincides with withdrawal.

2.5 Research gap

Prior work largely assumes that reputation and gossip raise cooperation, or at least do not systematically reduce it. LLM-agent papers that celebrate communication rarely measure post-stigma intensity under climate asymmetry across independent replications. Climate-finance simulations rarely instrument social stigma with pairwise consistency scoring and gossip bulletins. The gap is therefore joint. We need an environment where social marks, forced institutions, shocks, and redistribution coexist, and where multi-seed evidence distinguishes robust associations from seed-specific paths. The present study addresses that gap with a dual-run event study of contribution intensity after reputation and gossip events.

3. Methodology

3.1 Environment overview

We study a repeated public-goods economy with heterogeneous country roles. Each agent is labeled developed or developing. Labels govern initial wealth, vulnerability, and institutional assignment in the climate package. Interaction proceeds in discrete rounds. Within each round, agents choose

Stage-1 contributions inside their assigned institution. Agents in the enforcing institution then allocate Stage-2 sanctions and rewards. Climatic shocks, fund accounting, social scoring, and democracy occur on a fixed schedule described in Section 4.

Two institutions operate in parallel. The enforcing institution (SI) combines Stage-1 public-goods returns with peer punishment and reward. The non-enforcing institution (SFI) provides Stage-1 returns without Stage-2 sanctioning. Institution labels are presented to agents in climate language as a binding treaty versus a non-binding agreement. In the primary climate runs, assignment is forced every round. Developed agents enter SI. Developing agents enter SFI. Free institutional choice is reserved for boundary baselines discussed later.

3.2 Stage-1 public good

Let c_i denote agent i 's Stage-1 contribution and let G denote the set of agents in the same institution. Total contribution is $C = \sum_{j \in G} c_j$. With public-good multiplier m and group size $n = |G|$, each member receives Stage-1 share

$$\text{share}_i = \frac{m \cdot C}{n}. \quad (1)$$

The marginal per-capita return is therefore m/n . Contribution budgets equal current wealth, floored at zero. Agents receive decision cards that state budget bounds, group size, marginal return, and recent same-institution averages.

Contribution intensity is the primary behavioral outcome. For each agent-round we define

$$\text{prop}_{i,t} = \frac{c_{i,t}}{w_{i,t}}, \quad (2)$$

where $w_{i,t}$ is end-of-round wealth. Absolute contributions differ by orders of magnitude across groups. Intensity normalizes for scale and supports cross-agent event studies.

3.3 Stage-2 enforcement

Only SI members enter Stage-2. Each receives a sanction budget proportional to wealth, with a climate floor. Punishment tokens reduce a target's payoff at a fixed effect-to-cost ratio. Reward tokens raise a target's payoff at unit cost. Targets are listed with current contributions and pairwise consistency scores when available. SFI members skip Stage-2 entirely. Enforcement is itself a second-order public good. We track sanction burden as context rather than as the primary estimand.

3.4 Dual-use loss-and-damage fund

Stage-1 contributions are dual-use deposits. Each contributed unit also enters a collective loss-and-damage pool. The pool is hidden from agents as a live balance. Agents are told that contributions support both emissions-reduction framing and disaster payouts. After climatic shocks, developing agents may receive payouts subject to coverage caps and equity weights. Developed agents absorb damage without fund receipts under the default disbursement rule.

Gross damage for agent i after shock severity s is proportional to s and vulnerability v_i . Net damage equals gross damage minus any fund receipt. Coverage is the ratio of aggregate

payouts to aggregate gross damage in a shock round. High coverage can coexist with widening wealth gaps when collection and growth dominate disbursement.

3.5 Social information channels

Three linked channels create social pressure.

Pairwise consistency scoring. After contributions, evaluators score targets on a one-to-ten scale. High scores indicate match between stated intent and action. Low scores indicate inconsistency. Scores are elicited with short natural-language justifications.

Aggregate reputation. Agents observe an aggregate peer-trust rating derived from pairwise scores. Bad reputation is defined analytically as an aggregate score below four. A reputation drop is a decline of at least one point between rounds.

Gossip bulletins. Low pairwise scores generate gossip items injected into subsequent prompts. Bulletins name the observer, the target, the consistency score, and a short comment. Capacity limits keep only a top set of items. In the event study, gossip targeting is reconstructed from low-score conditions consistent with bulletin generation. The protocol does not retain a separate verbatim gossip transcript for every edge.

These channels are informational. They do not mechanically force higher contributions. Any repair must arise from prompted choice.

3.6 Democracy

Every five rounds, agents may propose and vote on a bounded whitelist of parameters. Whitelist items include subsidy fraction, subsidy breadth, punishment and reward effects, and loss-and-damage equity or damage weights. Majority outcomes update the live rule set. Democracy is endogenous institutional adaptation. It is analyzed as backdrop for contribution dynamics and as a seed-sensitive political path.

3.7 Agent decision interface

Agents are prompted language models with a default self-interested persona. They maximize cumulative payoff unless a specialized persona is assigned. Climate role guidance soft-frames developed versus developing constraints. Decision outputs are structured as contribution amounts, sanction maps, proposals, votes, and rationales. Belief states update after rounds with trust labels and institutional strategies. We analyze both numeric choices and coded language. We never equate fluent rationales with internalized norms.

Primary analyses treat each seed as a complete society under the same scientific protocol. Inference emphasizes sign agreement across seeds for the focal social-pressure association. Magnitude estimates from either seed alone remain uncertain because event cells are noisy and agents are interdependent within rounds.

4. Experimental Design

4.1 Primary multi-seed protocol

The primary evidence comprises two independent full-protocol runs that differ only by random seed. Each run uses Meta Llama 3.1 8B with temperature 0.5 and top $p = 0.95$ for ordinary decisions, lower temperatures for pairwise scoring and voting, and a 4096-token context window. Each society contains 26 agents (12 developed, 14 developing) interacting for 30 rounds. Institutional assignment is forced every round. Developed agents enter SI. Developing agents enter SFI. Public-good multiplier m equals 1.6. Climatic shocks occur in rounds 5 and 10 with severities 0.10 and 0.20. Democracy sessions occur every five rounds and therefore coincide with both shocks. Vulnerability is 0.30 for developed agents and 1.50 for developing agents. Stage-2 budgets in SI equal $\max(20, \lfloor 0.05 \times \text{wealth} \rfloor)$. Default punishment effect-to-cost is 3; reward effect-to-cost is 1. Gossip bulletins are eligible when pairwise scores are at most 7, with a capacity of five items.

Both runs are analyzed with equal weight. We do not designate either seed as primary and the other as supplemental. Where signs agree, we treat the association as robust under this protocol. Where signs diverge, we report heterogeneity rather than pooling into a single universal effect. Agent identifiers are not treated as fixed personas across seeds, because group labels reshuffle for a substantial subset of identifiers.

4.2 Outcomes and event definitions

The primary outcome is $\text{prop}_{i,t}$. Immediate event-study changes use $\Delta \text{prop}_{i,t} = \text{prop}_{i,t+1} - \text{prop}_{i,t}$. Bad reputation uses aggregate peer-trust below 4. Reputation drops use declines of at least 1.0. Gossip targets follow reconstructed eligibility consistent with the bulletin rule. Diff-in-diff contrasts, when reported for a seed, compare affected agents with same-round co-institution controls without the focal event.

Secondary outcomes include zero-contribution shares, round-mean trajectories by institution, Mann–Kendall trend tests on round means, developed–developing wealth gaps, loss-and-damage coverage ratios, adopted democratic rules, and coded motifs in post-event rationales. SI versus SFI mean comparisons are descriptive. Because institution membership is collinear with group label under forced routing, those comparisons are not causal institution effects.

4.3 Qualitative coding

For the first seed, contribution rationales in rounds immediately after bad-reputation or gossip events are coded with a multi-label codebook. Opportunistic covers free-riding, self-interest, payoff maximization, or marginal-return language. Reputation management covers explicit standing, credibility, or repair language. Conformity covers peer-average imitation. A second coder independently labels a stratified subsample for reliability. Cross-seed language dialects are summarized with keyness contrasts between SI and SFI contribution rationales.

4.4 Boundary baselines

Shock-free free-choice baselines from related abstract public-goods runs motivate the climate design. In those baselines, rep-

Table 1: Cooperation and inequality across two independent full-protocol seeds (780 agent-rounds each).

Statistic	Seed 1	Seed 2
Mean intensity (all)	0.293	0.365
SI / SFI mean intensity	0.291 / 0.296	0.364 / 0.366
SI / SFI median intensity	0.194 / 0.034	0.205 / 0.179
SI / SFI zero share	6.4% / 16.2%	11.1% / 22.4%
Agent-mean MW p (SI vs SFI)	0.817	0.777
Final wealth gap (dev. — developing)	2.15×10^8	5.83×10^8
Final LDF pool	4.34×10^9	1.30×10^{10}
Shock coverage (R5, R10)	0.768	0.768
Mann–Kendall trend on round means	NS	NS

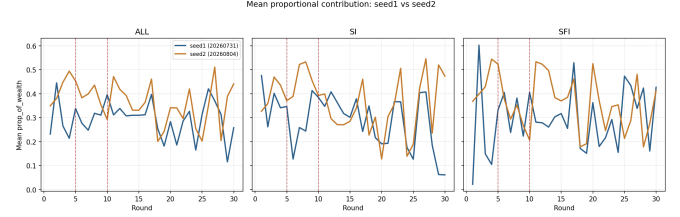


Figure 1: Round-mean contribution intensity by institution in two independent seeds. Seed 2 exhibits higher levels. Within-seed SI and SFI means remain close throughout.

utation can pull agents toward SI when exit is available, and greedy zero contribution dominates without existential risk. We cite those patterns only as boundary conditions. They are not re-estimated under the forced climate protocol and are not pooled with the two climate seeds.

5. Results

5.1 Cooperation levels across seeds

Table 1 summarizes both full-protocol societies. Mean contribution intensity is 0.293 in Seed 1 and 0.365 in Seed 2. Within each seed, SI and SFI means are nearly identical. Agent-mean Mann–Whitney tests do not reject equality of institution means ($p=0.817$ and $p=0.777$). Medians remain lower than means, especially for SFI in Seed 1 (0.034 versus mean 0.296). Seed 2 compresses that median gap (0.205 versus 0.179), yet still shows a thicker SFI zero share (22.4% versus 11.1% in SI). Average cooperation is therefore moderately positive in both societies, while distributions remain right-skewed.

Figure 1 overlays round-mean trajectories by institution across seeds. Seed 2 sits above Seed 1 for most of the horizon. Pearson correlation of round means across seeds is only 0.194 ($p=0.305$). Institution-specific correlations are likewise weak (SI $r=0.144$; SFI $r=0.020$). Mann–Kendall tests find no significant monotonic trend in either seed. The level shift is a seed-level cooperation increase rather than SI–SFI divergence. Because institution membership is forced by group, mean parity is descriptive rather than a causal institution effect.

Early-round composition differs sharply. In Seed 1, 71.4% of SFI agents contribute nothing in round 1, then SFI mean intensity jumps from 0.022 to 0.602 in round 2 once peer history appears. In Seed 2, SFI round 1 zeros are near zero and the cold-start escalation is muted. Figure 2 shows zero frequencies for Seed 1 as the clearer cold-start illustration. The contrast warns against treating any single opening path as structural. Both societies still converge to matched institution means after the opening window.

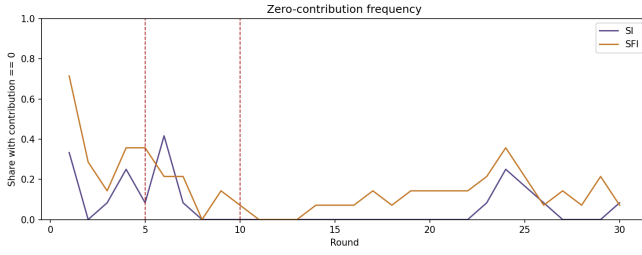


Figure 2: Share of agents contributing zero by institution and round in Seed 1. SFI opens with widespread zeros. SI shows a sharp zero spike after the first climatic shock.

Table 2: Immediate mean Δprop after social-information events by seed and institution. Negative cells indicate average withdrawal after the mark.

Event	Inst.	Seed 1	Seed 2
Bad reputation	SI	-0.036	+0.055
Bad reputation	SFI	-0.103	-0.069
Reputation drop	SI	-0.091	+0.058
Reputation drop	SFI	-0.119	-0.017
Gossip target	SI	-0.048	+0.187
Gossip target	SFI	-0.201	-0.084

5.2 Main result: social pressure by institution and seed

Table 2 and Figure 3 report immediate mean Δprop after social-information events. In both seeds, SFI cells are negative for bad reputation and gossip targeting. Gossip associations are -0.201 (Seed 1) and -0.084 (Seed 2). Bad-reputation associations are -0.103 and -0.069 . Reputation-drop associations in SFI remain negative, though attenuated in Seed 2. The developing/non-enforcing withdrawal pattern is the robust core of the study.

SI cells do not replicate in sign. In Seed 1, SI means after bad reputation, reputation drops, and gossip are negative (-0.036 , -0.091 , -0.048). In Seed 2, the same three means are positive ($+0.055$, $+0.058$, $+0.187$). A universal claim that gossip lowers intensity for all agents is therefore unsupported. The supported claim is institutional heterogeneity. Social marks associate with withdrawal among SFI agents across seeds, while SI responses are seed-sensitive.

Figure 4 shows the Seed 1 distribution of immediate changes. Mass lies below zero, especially for gossip targets in SFI. Cell-wise bootstrap intervals in Seed 1 are wide relative to the means, and standardized effects versus zero are small. The multi-seed contribution is therefore the sign pattern across replications rather than precise magnitude estimation from one seed alone.

For Seed 1, Table 3 contrasts affected agents with same-round controls. Diff-in-diff estimates are negative for all three event families. Permutation tests reject the null for bad reputation and gossip at conventional levels ($p=0.015$ and $p=0.039$). Controls show near-zero or slightly positive mean changes. The Seed 1 association is concentrated among marked agents rather than a global round effect. First events coincide with larger declines than repeats in Seed 1, consistent with attenuation without sign reversal. Cross-seed inference therefore privileges the SFI sign pattern over any single-seed universal DiD.

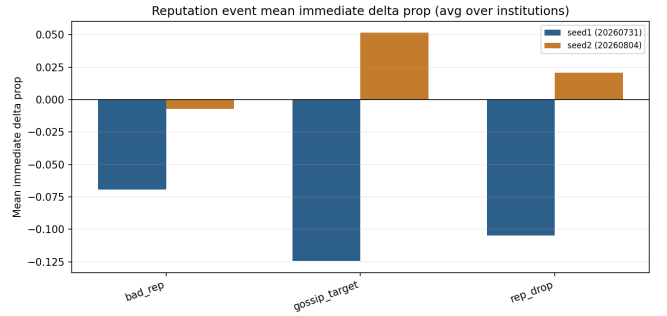


Figure 3: Immediate mean Δprop after social-information events by institution and seed. SFI associations remain negative. SI associations reverse sign between seeds.

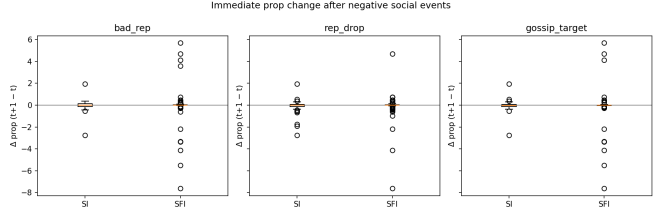


Figure 4: Distribution of immediate Δprop after social-information events in Seed 1. Mass lies below zero, especially for gossip targets in the non-enforcing institution.

Gossip targeting is uneven across agents in both societies. A minority of agents account for many bulletin appearances, consistent with ambient low pairwise scores. About 84% of consistency scores are at most seven in Seed 1 and about 90% in Seed 2, so gossip eligibility is widespread even when naming concentrates. Named agents are often relatively high contributors at the hit, so stigma does not land only on obvious free-riders.

5.3 Rationales after social events

Seed 1 post-event contribution texts ($N=114$) are coded with the multi-label codebook in Table 4. Opportunistic framing appears 29 times. Reputation-management language appears 3 times. Conformity appears 5 times. Explicit gossip references appear 0 times in the coded sample. Inter-coder agreement on a stratified subsample is substantial (percent agreement 0.83; Cohen's $\kappa=0.72$ for primary labels; $\kappa=0.91$ for opportunistic versus other). Agents seldom state that they will restore standing by giving more. They more often cite weak marginal returns and peer free-riding logic.

Illustrative paraphrases match the counts. One SFI agent, after observing high peer giving, chooses zero to conserve resources. Another SI agent after the first shock selects zero under a high-budget, low-marginal-return logic. A late-round SFI rationale seeks the minimum contribution that avoids a free-rider label. That last case is reputation-aware. It aims at a floor rather than at repair-driven escalation.

Cross-seed keyness contrasts reproduce a dialect split. SI contribution rationales emphasize strategy and self-interest terms. SFI rationales emphasize incentives and immediacy. Figure 5 shows Seed 1 contribution lexicons by institution. Fairness language is rare in shared contribution talk in both societies. Positive average transfers therefore coexist with weak

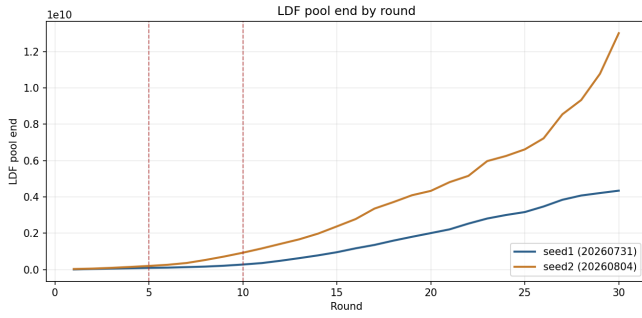


Figure 6: Loss-and-damage pool balances by round across seeds. Pools grow far beyond cumulative shock payouts.

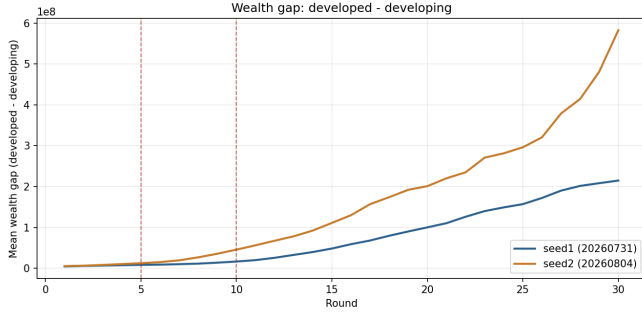


Figure 7: Developed-minus-developing mean wealth gap over rounds in both seeds. Gaps widen despite identical shock coverage ratios.

Positive average cooperation leaves the repair hypothesis unsupported. Both seeds sustain moderately positive mean intensity without strong path stability or fairness-heavy contribution talk. Seed 2’s higher mean intensity even coincides with a larger terminal wealth gap. The pattern matches warnings that LLM reasoning can sustain calculated free-riding alongside occasional giving (Li and Shirado 2025; Guzman Piedrahita et al. 2025). It also aligns with limited-norm findings in multi-agent LLM studies (Gupta et al. 2025; Ashery et al. 2025). Transfers are real. Internalized contribution norms are not established by average levels alone.

Forced assignment is essential to interpretation. Because developed agents cannot exit SI and developing agents cannot enter Stage-2, social marks cannot be answered by institutional flight of the kind observed in free-choice baselines. Withdrawal becomes a residual response for SFI agents who are named but cannot punish. Identification without sanction reach, already visible in mixed baselines, reappears here as stigma without repair. Ambient low consistency scores make gossip eligibility common, yet repair language remains rare. Naming is therefore cheap relative to behavioral correction.

Democratic adaptation adds a second layer of contingency. Both seeds expand subsidies and equity weights, yet only Seed 2 weakens punishment. Governance is endogenous and path dependent under identical whitelists. That divergence cautions against treating any one political trajectory as the climate society’s institutional destiny. It also confounds shock attribution, because votes and damages share a calendar. Costless meta-governance can substitute for costly peer enforcement without guaranteeing cooperative norms.

Fiscal outcomes reinforce the political-economy punchline. Identical coverage ratios coexist with different terminal pools and larger wealth gaps in the higher-cooperation seed. Collection is not equalization. A hidden dual-use pool can grow while stock inequality widens. Mechanism laboratories of this kind therefore speak to design tensions in climate risk sharing without evaluating any real-world fund’s effectiveness. The practical implication for LLM multi-agent design is narrower and sharper. Social information modules should be evaluated with event studies and multi-seed sign checks, especially when agents lack exit and Stage-2 reach.

7. Limitations

Several limits bound the claims. Only two seeds and one open-weight 8B instruct model are studied. Additional seeds are required before SI gossip responses can be declared unstable in a distributional sense rather than merely flipped once. Forced institutional assignment precludes causal SI-versus-SFI identification. Shock and democracy rounds coincide, so climate-damage effects are confounded with rule politics. Gossip targeting is reconstructed from eligibility rules rather than from a complete verbatim bulletin archive. Agent identifiers are not stable personas across seeds.

The environment is a stylized mechanism laboratory. It is not a calibrated model of Parties, pledges, or disbursement under the Fund for responding to Loss and Damage. Ostromian design principles are not fully implemented. Positive mean cooperation must not be read as proof of internalized norms. Boundary free-choice baselines motivate the climate package but are not re-run under the same multi-seed protocol here. External validity beyond this prompt family, model, and parameter whitelist remains open.

8. Conclusion

Reputation and gossip are often assumed to stabilize cooperation in LLM agent societies. Under climate risk, forced institutional asymmetry, and a dual-use loss-and-damage pool, that assumption fails for developing agents in the non-enforcing institution. Across two independent full-protocol seeds, social marks associate with lower next-round contribution intensity in SFI. Responses among enforcing-institution agents reverse sign across seeds. Average cooperation remains moderately positive, democratic adaptation is path dependent, and high shock coverage coexists with widening wealth gaps.

The nugget is institutional heterogeneity in social pressure. Monitoring without enforceable repair can coincide with withdrawal rather than with restored giving. Multi-seed evidence is what makes that claim credible. Future work should expand seeds and models, desynchronize shocks from democracy, restore free institutional choice under climate prompts, and ablate gossip thresholds. Those extensions can test whether restoring exit or Stage-2 reach converts social stigma back into cooperation technology.

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A. Appendix: Protocol, Prompts, and Supplementary Analyses

This appendix supports replication of the scientific protocol. It describes parameters, decision interfaces, and supplementary multi-seed results. Software-engineering artifacts are omitted.

A.1 Experimental configuration

Table 6 lists core configuration values shared by both climate-risk societies.

Table 6: Shared configuration for the two full-protocol seeds (scientific parameters).

Parameter	Value
Population	26 agents (12 developed, 14 developing)
Horizon	30 rounds
Institutional assignment	Forced: developed $\rightarrow$ SI; developing $\rightarrow$ SFI
Public-good multiplier m	1.6
Stage-1 budget	Current wealth (non-negative)
Stage-2 budget (SI)	$\max(20, \lfloor 0.05 \times \text{wealth} \rfloor)$
Punishment effect / cost (default)	3 / 1
Reward effect / cost	1 / 1
Initial subsidy fraction	0.20 of punishment costs to top contributors
Shock schedule	Round 5 severity 0.10; round 10 severity 0.20
Vulnerability	Developed 0.30; developing 1.50
Damage scale factor	$150000 \times \text{severity} \times \text{vulnerability}$
LDF deposit rule	Stage-1 contribution dual-use deposit each round
LDF recipients	Developing agents after shocks
Initial max coverage	0.90
Initial equity weight	0.00 (democratically revisable)
Democracy interval	Every 5 rounds
Consistency / gossip	Enabled; bulletin if score ≤ 7 ; max 5 items
Decision model	Meta Llama 3.1 8B
Default sampling	temperature 0.5, top $p = 0.95$
ToM / democracy temps	0.3 / proposal 0.5, vote 0.3
Context length	4096 tokens
Independent replications	Two seeds (Seed 1, Seed 2)

Currency units are presented to agents as million-scale fiscal units in a climate summit framing. Initial endowments follow group-specific schedules that create large developed–developing stock gaps at opening.

A.2 Round protocol

Each round proceeds in the following scientific order.

1. Institutional assignment is applied (forced in the primary runs).
2. Agents choose Stage-1 contributions under decision cards, climate role guidance, beliefs, and any gossip bulletins.
3. Stage-1 public-good returns are distributed within each institution.
4. Dual-use deposits update the loss-and-damage pool.
5. SI agents allocate Stage-2 punishments and rewards.
6. Pairwise consistency scoring updates peer judgments and aggregate reputation.
7. Eligible low scores generate gossip bulletins for later rounds.
8. If the round index is divisible by five, agents propose and vote on whitelist parameters.
9. If the round is a shock round, damages and developing-country payouts are realized.
10. Wealth, beliefs, and histories update for the next round.

A.3 Outcome construction

Contribution intensity uses end-of-round wealth as denominator. Immediate event-study changes use $\text{prop}_{t+1} - \text{prop}_t$. Bad reputation uses aggregate peer-trust below 4. Reputation drops use declines of at least 1.0. Gossip targets follow reconstructed eligibility consistent with the bulletin rule. Diff-in-diff controls are co-institution agents in the same round without the focal event. Cross-seed comparisons use identical outcome definitions on both replications.

A.4 Prompt templates (sanitized)

Prompts are shown as scientific templates with placeholders. Numeric inventories and peer lists are omitted.

A.4.1 Default behavioral persona

Act in whatever way maximizes your own cumulative payoff over the game. Cooperation, free-riding, punishment, or defection are all valid if they serve your interests. Do not assume you have any obligation to the group. Treat other agents as strategic variables. Use past rounds to identify patterns you can exploit or respond to. Keep choices valid under all game rules and cite the concrete facts driving your decision.

A.4.2 Climate role guidance

Developed country. You typically have higher fiscal capacity and historically higher emissions. Treat long-run climate stability and institutional credibility as strategically important. You are assigned to the binding treaty institution with peer sanctioning.

Developing country. You typically face higher climate vulnerability and tighter fiscal constraints. Treat disaster compensation and survival constraints as strategically important. You are assigned to the non-binding agreement without peer sanctioning.

A.4.3 Stage-1 contribution decision

Choose an integer contribution between 0 and your current budget. You observe group size, marginal per-capita return, recent same-institution averages, own wealth, beliefs, and any gossip bulletins. State a brief rationale for the chosen amount.

A.4.4 Stage-2 sanction decision (SI only)

Allocate your sanction budget across listed peers as punishments and rewards. Punishments and rewards must be non-negative integers that respect the budget. Provide a brief justification that is consistent with the numeric allocation.

A.4.5 Pairwise consistency scoring

Score the target from 1 to 10 for consistency between stated intent and observed action. Provide a one-sentence justification. Low scores may generate gossip bulletins.

A.4.6 Democracy proposal and vote

Propose at most one whitelist parameter change, or abstain. Later, vote yes or no on each eligible proposal. Whitelist items include subsidy fraction, subsidy breadth, punishment effect, reward effect, and loss-and-damage equity or damage weights.

A.5 Supplementary multi-seed statistics

Table 7 reports cross-seed correlations of round-mean intensity. Correlations are weak for the pooled series and near zero within institutions. Per-agent mean-intensity correlations across seeds are near zero even among identifiers that keep the same group label, confirming that agent identifiers are not stable personas. Table 8 reports Wilcoxon tests for within-agent post-minus-pre intensity around shock rounds. Figure 8 shows shock-delta distributions by seed. Figure 9 shows the agent-mean intensity scatter across seeds. Figure 10 shows shared-prompt concept rates for Seed 1 as additional lexical context.

Table 7: Cross-seed correlation of round-mean contribution intensity.

Series	Pearson r	p	RMSE
All agents	0.194	0.305	0.124
SI only	0.144	0.448	0.164
SFI only	0.020	0.916	0.184

Table 8: Within-agent shock event-study Wilcoxon tests for Δprop .

Seed	Shock	Mean Δ	Median Δ	Wilcoxon p
1	R5	+0.023	−0.014	0.367
1	R10	+0.010	+0.015	0.328
2	R5	−0.080	−0.011	0.582
2	R10	+0.050	−0.147	0.367

A.6 Motif codebook definitions

- *Opportunistic*: free-riding, self-interest, payoff maximization, or marginal-return language as the stated driver.
- *Conformity*: peer or group-average behavior as the reason for the agent’s own choice.
- *Reputation management*: explicit standing, credibility, trust scores, or avoiding a free-rider label.
- *Named agents*: reference to a specific peer identifier.
- *Future-rounds*: intertemporal conservation without a stronger primary motif.
- *Gossip reference*: explicit gossip or bulletin language.



Figure 8: Distribution of within-agent post-minus-pre intensity changes around climatic shocks, by seed.

A.7 Claim boundaries

The following claims are supported under the present protocol: SFI withdrawal after gossip/bad reputation across seeds; within-seed $SI \approx SFI$ mean intensity; coverage without equalization; path-dependent democracy. The following claims are not supported: causal SI-versus-SFI institution effects; universal SI gossip withdrawal; evaluation of any real-world loss-and-damage fund; full implementation of Ostrom design principles; persona-stable agent identifiers across seeds.

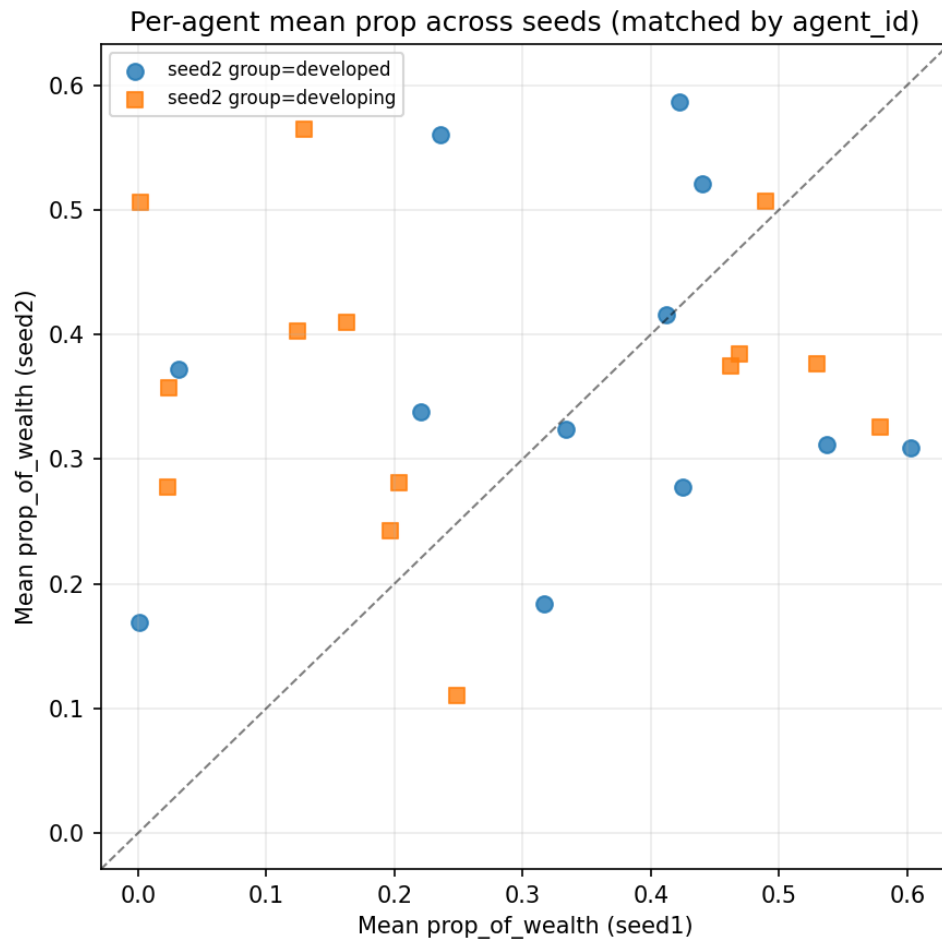


Figure 9: Agent-mean contribution intensity in Seed 1 versus Seed 2. Points are not persona-matched across seeds.

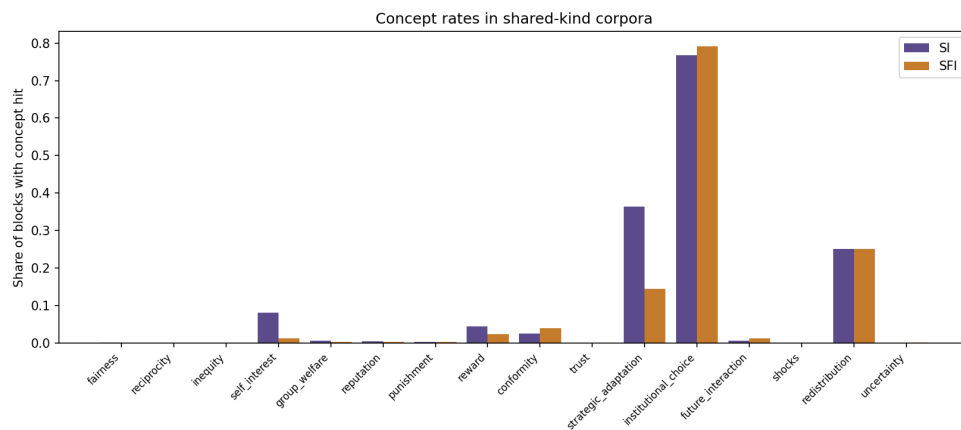


Figure 10: Concept rates in shared rationales (Seed 1). Fairness and reciprocity remain rare relative to strategy language.